Huawei Access Node HEB-ACC-75 Hebbal Edge Optical Transport Failure SOP (Minor Severity)

Node: HEB-ACC-75 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_8123

Description

This Standard Operating Procedure (SOP) outlines the steps to diagnose and resolve an Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) on Huawei Access Node HEB-ACC-75 at Hebbal Edge. The failure domain is primarily attributed to the degradation of SFP/XFP modules or fiber cuts that disrupt the optical signal transmission between the node and the neighboring nodes. The Huawei Access Node employs Optical Transport Network (OTN) technology, which utilizes the G.709 standard for wavelength division multiplexing (WDM) and the ITU-T G.6941 standard for Wavelength Division Multiplexing (WDM) transmission. OTN frames are encapsulated using Optical Channel Transport Units (OTU) and transmitted over optical fibers. The Access Node acts as a transparent bridge between the OTN layer and the higher layers, ensuring error-free transmission of client signals. However, degradation of SFP/XFP modules or fiber cuts can cause a loss of signal quality, resulting in Optical Transport Failure. The Severity of this failure is categorized as Minor, indicating minimal impact on network services.

Pre-requisite Checks

- pre_requisite: Verify that all necessary safety precautions have been taken before accessing the Huawei Access Node HEB-ACC-75 at Hebbal Edge.

- pre_requisite: Ensure that all necessary permissions and access credentials have been obtained to access the Huawei Access Node HEB-ACC-75 at Hebbal Edge.

- pre_requisite: Verify that the redundant link status is available and functional.

Potential Root Causes

Cause	Probability
SFP/XFP degradation	High
Fiber cuts	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

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Command: display fault-info, Expected Output: Fault information display, Description: This command displays the fault information of the Huawei Access Node HEB-ACC-75 at Hebbal Edge, including the fault type, location, and time. It assists in identifying the root cause of the Optical Transport Failure.
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Command: display otu-info, Expected Output: OTU information display, Description: This command displays the OTU information of the Huawei Access Node HEB-ACC-75 at Hebbal Edge, including the OTU status, transmit power, and receive power. It helps in diagnosing the Optical Transport Failure by checking the OTU status and power levels.

Command: display wdm-info, Expected Output: WDM information display, Description: This command displays the WDM information of the Huawei Access Node HEB-ACC-75 at Hebbal Edge, including the WDM status, channel number, and wavelength. It assists in identifying the specific WDM channel affected by the Optical Transport Failure.
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Resolution Steps

Step 1: Step: Replace the SFP/XFP module, Command: replace wdm-cwdm-sfp , Description: This step involves replacing the SFP/XFP module responsible for the Optical Transport Failure. The command 'replace wdm-cwdm-sfp' is used to replace the SFP/XFP module, and the should be specified accordingly.

Step 2: Step: Verify the fiber connection and repair if necessary, Command: display wdm-cwdm-fiber-info, Description: This step involves verifying the fiber connection between the Huawei Access Node HEB-ACC-75 at Hebbal Edge and the neighboring nodes. If the fiber connection is damaged, repair the fiber as necessary.

Step 3: Step: Adjust the Rx/Tx power, Command: set wdm-cwdm-power-adjust , Description: This step involves adjusting the Rx/Tx power level of the Huawei Access Node HEB-ACC-75 at Hebbal Edge to ensure optimal signal transmission. The command 'set wdm-cwdm-power-adjust' is used to adjust the Rx/Tx power level, and the should be specified accordingly.

Post-Check Verification

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Command: display otu-info, Expected Output: OTU information display, Description: This command is used to verify the OTU status and power levels after the resolution steps have been completed. It ensures that the Optical Transport Failure has been resolved.

Command: display wdm-info, Expected Output: WDM information display, Description: This command is used to verify the WDM status and channel number after the resolution steps have been completed. It ensures that the Optical Transport Failure has been resolved.
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Roll-back Procedures

- Step: Restore the original SFP/XFP module, Command: replace wdm-cwdm-sfp , Description: This step involves restoring the original SFP/XFP module that was replaced during the resolution steps. The command 'replace wdm-cwdm-sfp' is used to replace the SFP/XFP module, and the should be specified accordingly.

- Step: Verify the fiber connection and restore if necessary, Command: display wdm-cwdm-fiber-info, Description: This step involves verifying the fiber connection between the Huawei Access Node HEB-ACC-75 at Hebbal Edge and the neighboring nodes. If the fiber connection was repaired during the resolution steps, restore the original fiber connection.

- Step: Restore the original Rx/Tx power settings, Command: set wdm-cwdm-power-adjust , Description: This step involves restoring the original Rx/Tx power settings of the Huawei Access Node HEB-ACC-75 at Hebbal Edge. The command 'set wdm-cwdm-power-adjust' is used to adjust the Rx/Tx power level, and the should be specified accordingly.